C 40204 **(Pages : 3)** **Name.....**

Reg. No.....

SIXTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
MARCH 2023

Physics/Applied Physics

PHY 6B 11/APY 6B 12—SOLID STATE PHYSICS, SPECTROSCOPY AND LASER PHYSICS

(2017—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Section A

Answer in a word **or** *phrase*
Answer **all** *questions ; each question carries* 1 *mark.*

1. Transition temperature of mercury is ————————.
2. Which theory explains the phenomenon of superconductivity ?
3. Vector form of Bragg's law is ————————.
4. In ac Josephson effect current oscillates with ———————— frequency
5. Transition temperature Tc and critical field Hc for a superconductor are related as ————————.
6. The temperature at which a conductor become superconductor is ————————.
7. Give the expression for lattice constant '*a*' of a cubic system.
8. Give an example of highly coherent source of light.
9. Give an example of high temperature superconductor.
10. In hologram phase variation is recorded, True *or* False.

(10 × 1 = 10 marks)

Section B

Answer in a short paragraph-three **or** *four sentences.*
Answer **all** *questions ; each question carries* 2 *marks.*

11. Stocks or antistockes, which are more intense ? Why ?
12. What is optical pumping associated with LASER ?

Turn over

13. Sketch the temperature dependence of the electronic specific heat in the normal and superconducting states.

14. What is isotopic effect in superconductors ?

15. What is meant by second order phase transition ?

16. What is meant by atomic packing factor ?

17. What are miller indices ?

(7 × 2 = 14 marks)

Section C

Answer in **a paragraph** *of about* **half a page to one page**.
Answer any **five** *questions ; each question carries* 4 *marks.*

18. Discuss Type I and Type II superconductors.

19. Discuss rotational spectrum of rigid diatomic molecule.

20. Write a note on intensity of spectral transitions.

21. What are symmetry elements ?

22. Discuss quantum theory of Raman effect.

23. Discuss important characteristics of LASER.

24. Discuss power application of superconductors.

(5 × 4 = 20 marks)

Section D

Problems - write all relevant formulas. All important steps carry separate marks.
Answer any **four** *questions ; each question carries* 4 *marks.*

25. Find the miller indices of a plane that makes intercepts as a, b and c axes equal to 3A°, 4A° and 3A° in a tetragonal crystal with c/a ratio as 1.5.

26. Calculate the inter planar spacing for (321) plane in simple cubic lattice with inter atomic spacing a = 4.21 A°.

27. In the CO molecule the wave number difference between the successive absorption lines in the pure rotational spectrum is 384 m^{-1}. Calculate the moment of inertia of the molecule and the equilibrium bond length of the molecule. Masses of the C^{12} and O^{16} atoms are respectively 1.99×10^{-26} Kg and 2.66×10^{-26} Kg.

28. Calculate the critical current density for 1 mm diameter wire of lead at (a) 4.2K ; and (b) 7K. A parabolic dependence of He upon T may be assumed. Given Tc for lead is 7.18 K and H_0 for lead is 6.5×10^4 A/m.

29. The normal modes of vibration of CO_2 molecule are v_1 =1330 cm^{-1}, v_2 = 667 cm^{-1}, v_3 = 2349 cm^{-1}. Evaluate the zero point energy of CO_2 molecule.

30. Light of wavelength 1.5 μm incident on a material with a characteristic Raman frequency of 20×10^{12} Hz. Calculate the wavelength of stocks- shifted line (given $c = 3 \times 10^8$ m/s).

31. Calculate the number of photons emitted per second from a 1 Watt Ar - ion Laser operating at 488 nm.

(4 × 4 =16 marks)

Section E (Essays)

Answer in about **two pages**.

Answer any **two** *questions ; each question carries* 10 *marks.*

32. Discuss BCS theory of superconductors. What is Meissner effect ?

33. Explain the working of He-Ne Laser with energy level diagram.

34. Discuss vibrational Raman spectrum of symmetric top molecule.

35. Discuss vibrational rotational spectrum by considering diatomic vibrating rotator.

(2 × 10 = 20 marks)